THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — SCIENCE · HARD LEVEL CHAPTER TEST SET 3

Chapter 12: **Reproduction: How Life Continues** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Assertion (A): Menstruation does not occur during pregnancy. Reason (R): The uterine lining is retained to support the embryo. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

2. The ratio of the male to female babies born is roughly: 1

(a) 1 : 1 **(b)** 1 : 2 **(c)** 2 : 1 **(d)** 3 : 1

3. Plants grown by layering are produced from: 1

(a) branch while it is still attached to the parent
(b) seed
(c) spore
(d) leaf

4. Seeds can remain dormant for years and still germinate. This is possible because they have: 1

(a) low water content and food stored **(b)** no food
(c) no embryo **(d)** much water

5. The corpus luteum secretes: 1

(a) testosterone **(b)** progesterone **(c)** adrenaline **(d)** insulin

6. Assertion (A): Spores are produced in large numbers. Reason (R): Each spore has a high chance of finding a suitable place. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

7. A sperm with X chromosome fertilises the egg. The baby will be a: 1

(a) twin **(b)** girl **(c)** boy **(d)** neither

8. Vegetative propagation helps in producing plants that have lost the capacity to produce seeds, like: 1

(a) pea **(b)** banana **(c)** gram **(d)** wheat

9. **Assertion (A):** Animals that show external fertilisation, such as frogs, lay a large number of eggs. 1
Reason (R): Eggs fertilised outside the body are exposed to predators and other hazards, so few survive.
Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

10. **Assertion (A):** The sex of a human baby is determined by the sperm. 1
Reason (R): Every sperm carries a Y chromosome.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Why do mosses need water for reproduction? 2

12. What would be the effect of a mutation in a sexually reproducing organism and in an asexually reproducing one? 2

13. Why does the corpus luteum degenerate if fertilisation does not occur? 2

14. Why are the spores of the bread mould light and dry? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. State three differences between mitosis and meiosis, and where each occurs in the human body. 3

16. Explain the process of regeneration in Planaria and in humans (wound healing) and compare. 3

17. A seed has the following structures: seed coat, radicle, plumule, two cotyledons. Identify the type of seed and the fate of each structure. 3

OR

Explain how a sperm cell is adapted to its function and compare it with a white blood cell and a nerve cell. 3

18. Explain how the number of chromosomes is maintained from generation to generation in sexually reproducing organisms. 3

OR

Explain the significance of the fruit for the plant and for humans. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** A biology teacher explains human reproduction. A sperm fuses with an egg in the fallopian tube to form a zygote, which divides repeatedly to form an embryo. The embryo implants in the uterus and is nourished through the placenta for about nine months. 5
(i) What is the fusion of a sperm and an egg called?
(ii) Where does fertilisation take place?
(iii) What is the function of the placenta?
(iv) The zygote divides repeatedly. How many cells are present after 3 divisions, and what does the zygote carry in terms of the number of chromosomes?

20. Explain the events of the menstrual cycle in a 28-day cycle: the release of the egg, the changes in the uterine lining, and the events if fertilisation occurs or does not occur. 5

OR

Design a project on pollinators in your locality: select flowers, observe visits, record the data, analyse and suggest the conservation of pollinators. 5

END OF QUESTION PAPER — Total: 40 Marks